



Marwadi
University
Marwadi Chandarana Group



LABORATORY MANUAL

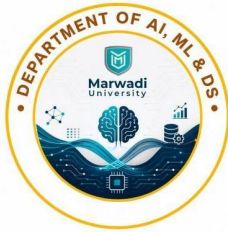
PROGRAMMING WITH PYTHON

Course Code: 01AP0303 | Credits: 4 | Practical: 2 Hours/Week

Student details	To be completed
Name	
Enrollment No.	
Semester / Class	
Laboratory Batch	
Academic Year	
Faculty	

Department of B.Tech CSE-AI/ML/DS-PBL
Faculty of Engineering and Technology
Marwadi University, Rajkot

Laboratory Record Certificate



Marwadi
University
Marwadi Chandarana Group



This is to certify that _____, Enrollment No. _____,

has satisfactorily completed the prescribed laboratory work for Programming with Python (01AP0303) during Academic Year
2026-27.

Role	Name / Signature	Date
Course Faculty		
Laboratory Faculty		
Head of Department		

Student Declaration

The work recorded in this manual has been performed and understood by the student. External code, data, APIs, libraries and AI-assisted contributions, if any, must be acknowledged. Credentials, personal data and unauthorized content must not be included in submitted files.

General Laboratory Instructions

- Attend the laboratory punctually and complete each experiment during the scheduled session.
- Create a separate folder for each experiment and use meaningful filenames.
- Write and test the program independently; do not submit code without understanding it.
- Use comments, meaningful identifiers, modular functions and appropriate exception handling.
- Record the algorithm, source code, test cases, output evidence and conclusion.
- Never store passwords, API keys or private personal data directly in source code.
- Maintain academic integrity and acknowledge external or AI-assisted material.
- Take a backup of code and data after every laboratory session.

GUI Output Screenshots

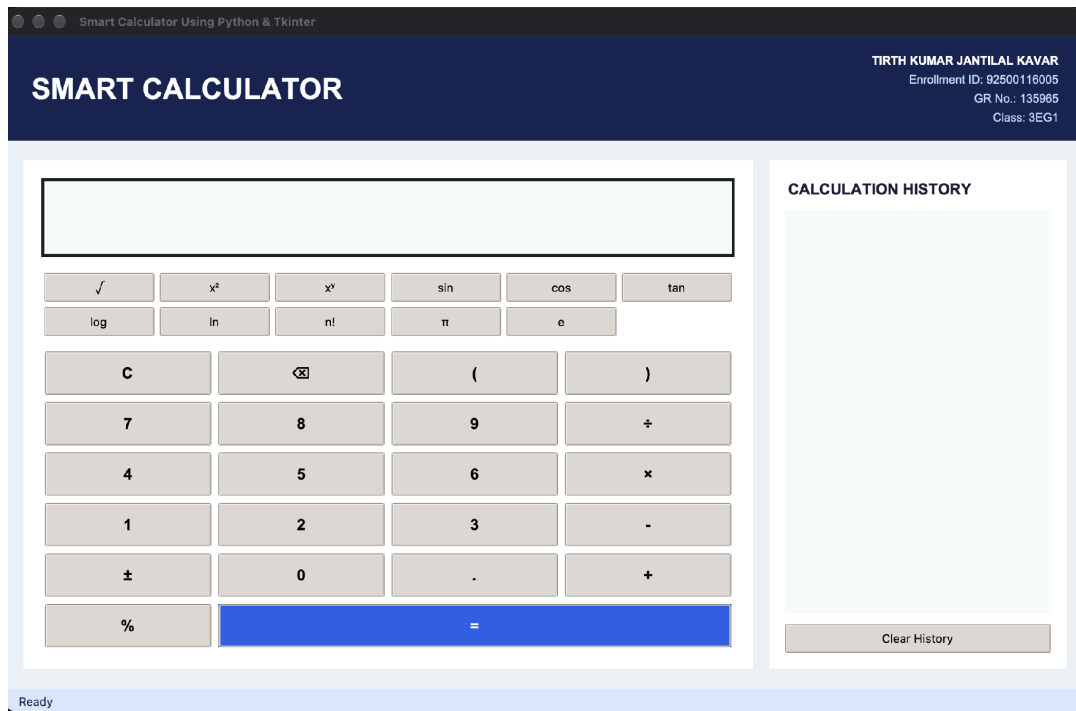


Figure 1.1 Initial GUI layout and calculator keypad interface

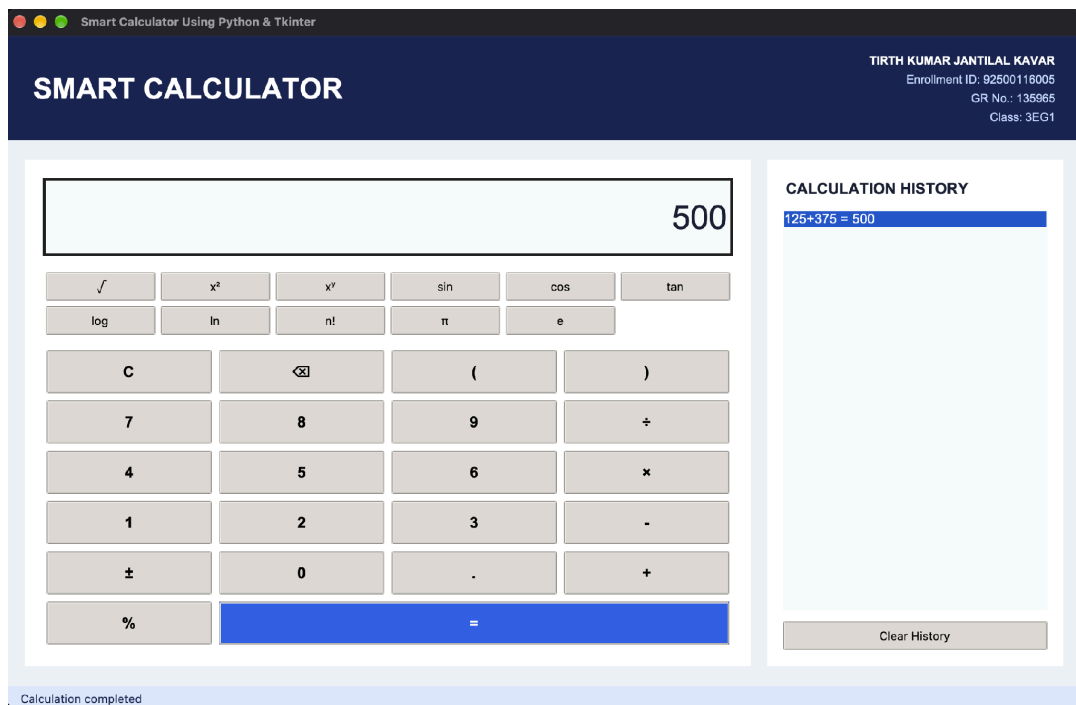


Figure 1.2 Arithmetic operation and standard expression evaluation

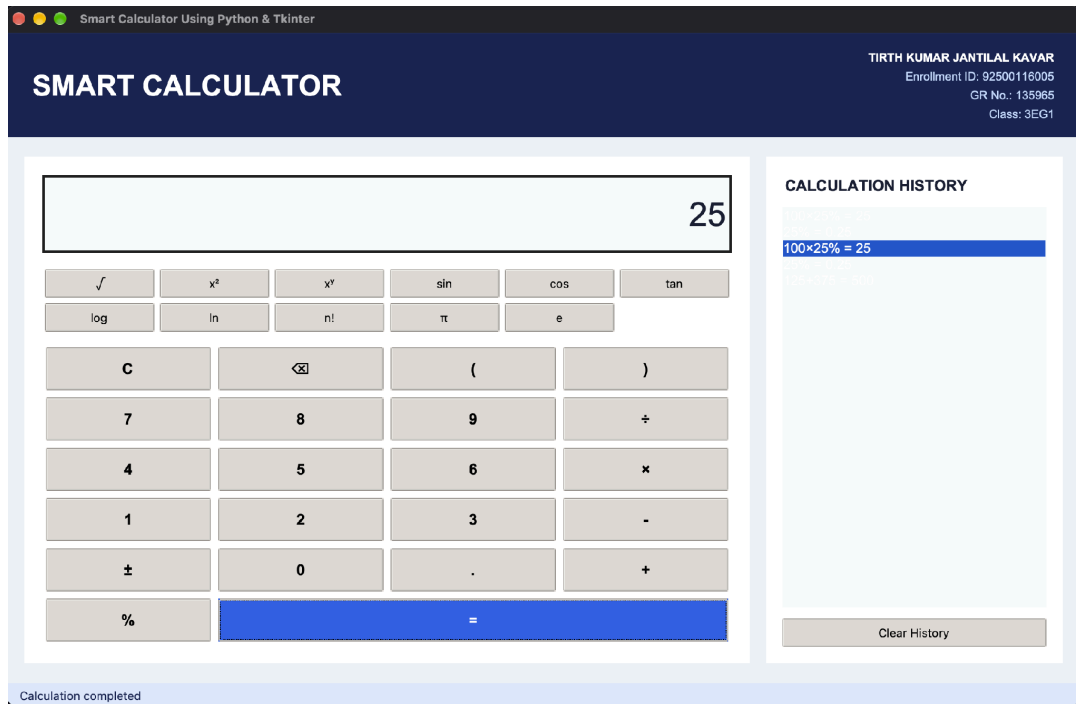


Figure 1.3 Percentage calculation and result display

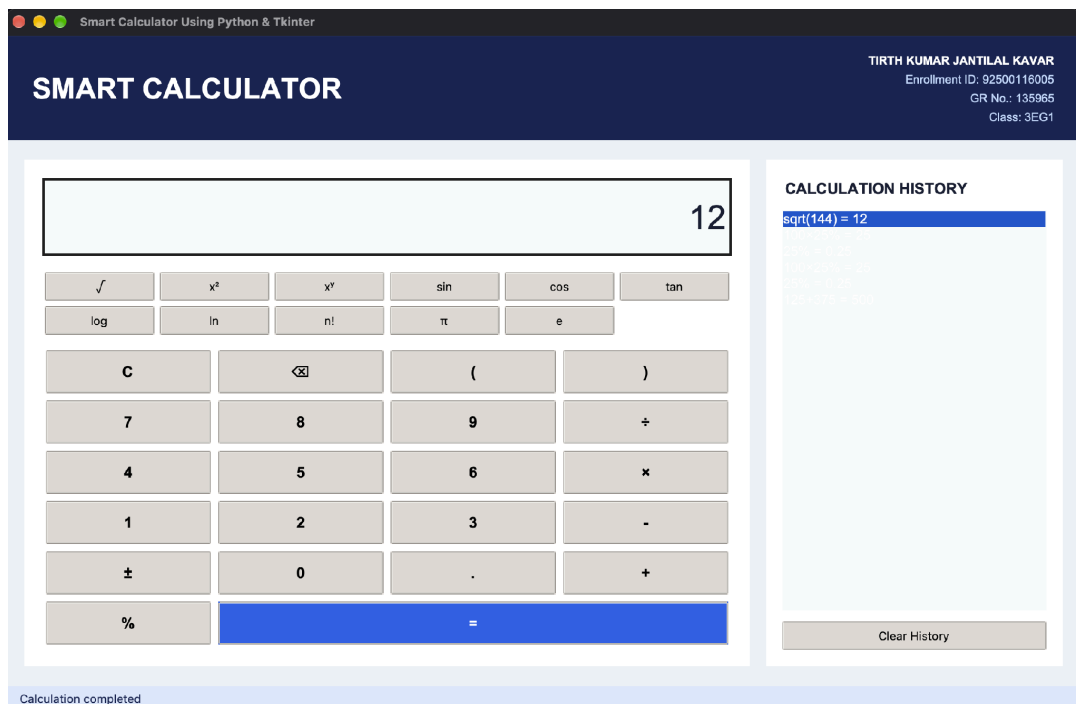


Figure 1.4 Scientific functions, roots, powers and constants

Smart Calculator Using Python & Tkinter

TIRTH KUMAR JANTILAL KAVAR

Enrollment ID: 92500116005

GR No.: 135965

Class: 3EG1

SMART CALCULATOR

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CALCULATION HISTORY

log(100) = 2

Clear History

Calculation completed

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GUI Output Screenshots

Employee Payroll System Using Python

EMPLOYEE PAYROLL SYSTEM

TIRTH KUMAR JANTILAL KAVAR
Enrollment ID: 92500118005
GR No.: 135985
Class: 3EG1

Employee Details

Employee ID E001	Employee Name Aarav Patel	Department IT	Basic Salary 30000
Working Days 26	Overtime Hours 10	Overtime Rate 200	Bonus 1500
Allowances 1000	Tax Percentage 10	Other Deductions 500	

Calculate Salary Add Employee Update Employee Delete Employee Clear Fields Export CSV

TOTAL EMPLOYEES 1	TOTAL GROSS PAYROLL ₹34,500.00	TOTAL TAX ₹3,450.00	TOTAL NET PAYROLL ₹30,550.00
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Employee ID	Name	Department	Basic Salary	Overtime	Gross Salary	Tax	Deductions	Net Salary
E001	Aarav Patel	IT	30000.00	2000.00	34500.00	3450.00	3950.00	30550.00

Ready

Figure 2.1 Payroll form layout and employee details entry fields

Employee Payroll System Using Python

EMPLOYEE PAYROLL SYSTEM

TIRTH KUMAR JANTILAL KAVAR
Enrollment ID: 92500118005
GR No.: 135985
Class: 3EG1

Employee Details

Employee ID E001	Employee Name Aarav Patel	Basic Salary 30000
Working Days 26	Overtime Hours 10	Bonus 1500
Allowances 1000	Tax Percentage 10	

Gross Salary: ₹34,500.00
Tax: ₹3,450.00
Total Deductions: ₹3,950.00
Net Salary: ₹30,550.00

OK

Calculate Salary Add Employee Update Employee Delete Employee Clear Fields Export CSV

TOTAL EMPLOYEES 1	TOTAL GROSS PAYROLL ₹34,500.00	TOTAL TAX ₹3,450.00	TOTAL NET PAYROLL ₹30,550.00
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Employee ID	Name	Department	Basic Salary	Overtime	Gross Salary	Tax	Deductions	Net Salary
E001	Aarav Patel	IT	30000.00	2000.00	34500.00	3450.00	3950.00	30550.00

Ready

Figure 2.2 Salary components, overtime rate and hours input

Employee Payroll System Using Python

EMPLOYEE PAYROLL SYSTEM

TIRTH KUMAR JANTILAL KAVAR
Enrollment ID: 92500116005
GR No.: 135985
Class: 3EG1

Employee Details

Employee ID: E001 Employee Name: Aarav Patel Basic Salary: 45000
Working Days: 26 Overtime Hours: 8 Gross Salary: ₹49,100.00
Allowances: 1000 Tax Percentage: 10 Tax: ₹4,810.00
Total Deductions: ₹5,410.00
Net Salary: ₹43,690.00

OK

Calculate Salary Add Employee Update Employee Delete Employee Clear Fields Export CSV

TOTAL EMPLOYEES: 1 TOTAL GROSS PAYROLL: ₹34,500.00 TOTAL TAX: ₹3,450.00 TOTAL NET PAYROLL: ₹30,550.00

Employee ID	Name	Department	Basic Salary	Overtime	Gross Salary	Tax	Deductions	Net Salary
E001	Aarav Patel	IT	30000.00	2000.00	34500.00	3450.00	3950.00	30550.00

Ready

Figure 2.3 Calculated gross salary, tax deduction and net pay summary

Employee Payroll System Using Python

EMPLOYEE PAYROLL SYSTEM

TIRTH KUMAR JANTILAL KAVAR
Enrollment ID: 92500116005
GR No.: 135985
Class: 3EG1

Employee Details

Employee ID: E001 Employee Name: Aarav Patel Basic Salary: 75000
Working Days: 26 Overtime Hours: 18 Gross Salary: ₹81,100.00
Allowances: 1000 Tax Percentage: 10 Tax: ₹8,110.00
Total Deductions: ₹8,610.00
Net Salary: ₹72,490.00

OK

Calculate Salary Add Employee Update Employee Delete Employee Clear Fields Export CSV

TOTAL EMPLOYEES: 1 TOTAL GROSS PAYROLL: ₹81,100.00 TOTAL TAX: ₹8,110.00 TOTAL NET PAYROLL: ₹72,490.00

Employee ID	Name	Department	Basic Salary	Overtime	Gross Salary	Tax	Deductions	Net Salary
E001	Aarav Patel	IT	75000.00	3600.00	81100.00	8110.00	8610.00	72490.00

Employee updated

Figure 2.4 Employee records table and payroll overview dashboard

Employee Payroll System Using Python

EMPLOYEE PAYROLL SYSTEM

TIRTH KUMAR JANTILAL KAVAR
Enrollment ID: 92500116005
GR No.: 135985
Class: 3EG1

Employee Details

Employee ID	Employee Name	Department	Basic Salary
E002	AGAMYA PANDEY	RELATIONS	200000
Working Days	Overtime Hours	Overtime Rate	Bonus
5	2	1000	30000
Allowances	Tax Percentage	Other Deductions	
0	18	0	

Calculate Salary Add Employee Update Employee Delete Employee Clear Fields Export CSV

TOTAL EMPLOYEES	TOTAL GROSS PAYROLL	TOTAL TAX	TOTAL NET PAYROLL
2	₹313,100.00	₹49,870.00	₹262,730.00

Employee ID	Name	Department	Basic Salary	Overtime	Gross Salary	Tax	Deductions	Net Salary
E001	Aarav Patel	IT	75000.00	3600.00	81100.00	8110.00	8610.00	72490.00
E002	AGAMYA PANDEY	RELATIONS	200000.00	2000.00	232000.00	41760.00	41760.00	190240.00

Employee added

Figure 2.5 Employee record deletion and CSV export verification

GUI Output Screenshots

Student Grade Analyzer Using Python

STUDENT GRADE ANALYZER

TIRTH KUMAR JANTILAL KAVAR
Enrollment ID: 92500116005
GR No.: 135965
Class: 3EG1

Student ID: S001 Student Name: Diya Shah Subject 1 Marks: 92 Subject 2 Marks: 88
Subject 3 Marks: 95 Subject 4 Marks: 90 Subject 5 Marks: 89

Calculate Result Add Student Update Student Delete Student Clear Export Report

TOTAL STUDENTS: 1 PASSED: 1 FAILED: 0 CLASS AVERAGE: 90.80% HIGHEST: 90.80% LOWEST: 90.80%

ID	Name	Sub 1	Sub 2	Sub 3	Sub 4	Sub 5	Total	%	Grade	Result
S001	Diya Shah	92	88	95	90	89	454	90.80	A+	Pass

Name: Diya Shah
Total Marks: 454 / 500
Percentage: 90.80%
Grade: A+
Result: Pass
Strongest Subject: Subject 3
Weakest Subject: Subject 2

Ready

Figure 3.1 Grade analyzer interface and marks entry form

Student Grade Analyzer Using Python

STUDENT GRADE ANALYZER

TIRTH KUMAR JANTILAL KAVAR
Enrollment ID: 92500116005
GR No.: 135965
Class: 3EG1

Student ID: S002 Student Name: AGAMYA PANDEY Subject 1 Marks: 90 Subject 2 Marks: 88
Subject 3 Marks: 92 Subject 4 Marks: 85 Subject 5 Marks: 95

Calculate Result Add Student Update Student Delete Student Clear Export Report

TOTAL STUDENTS: 2 PASSED: 2 FAILED: 0 CLASS AVERAGE: 90.40% HIGHEST: 90.80% LOWEST: 90.00%

ID	Name	Sub 1	Sub 2	Sub 3	Sub 4	Sub 5	Total	%	Grade	Result
S001	Diya Shah	92	88	95	90	89	454	90.80	A+	Pass
S002	AGAMYA PANDEY	90	88	92	85	95	450	90.00	A+	Pass

Name: AGAMYA PANDEY
Total Marks: 450 / 500
Percentage: 90.00%
Grade: A+
Result: Pass
Strongest Subject: Subject 5
Weakest Subject: Subject 4

Student added

Figure 3.2 Five-subject marks entry and boundary validation

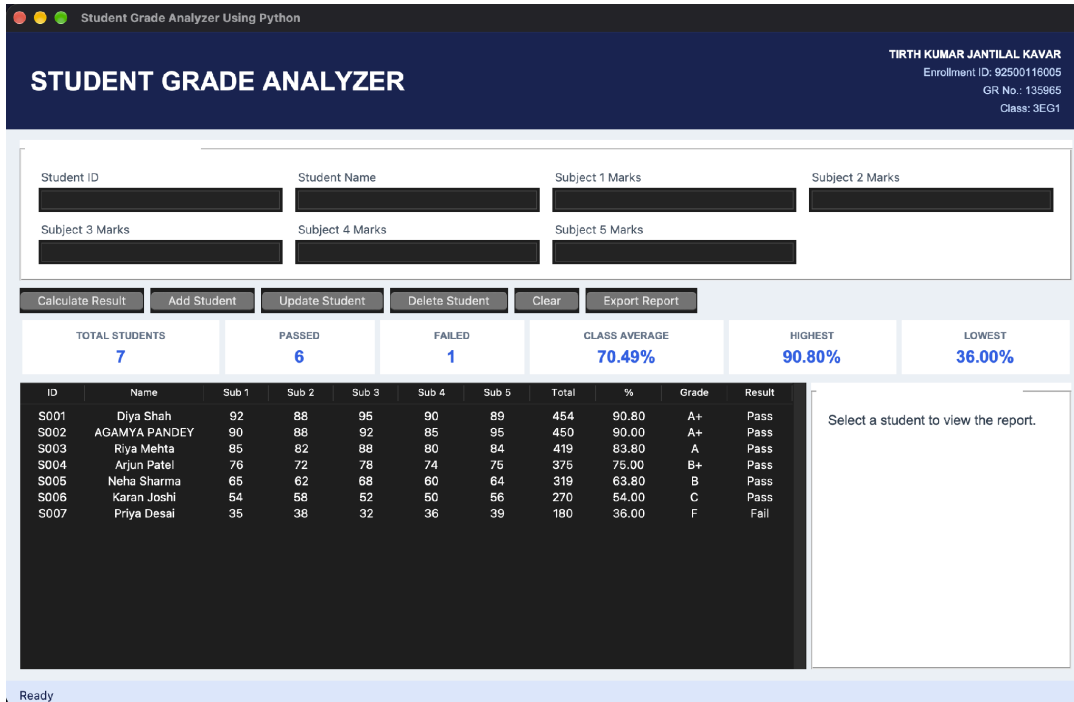


Figure 3.3 Computed percentage, grade, and pass/fail result

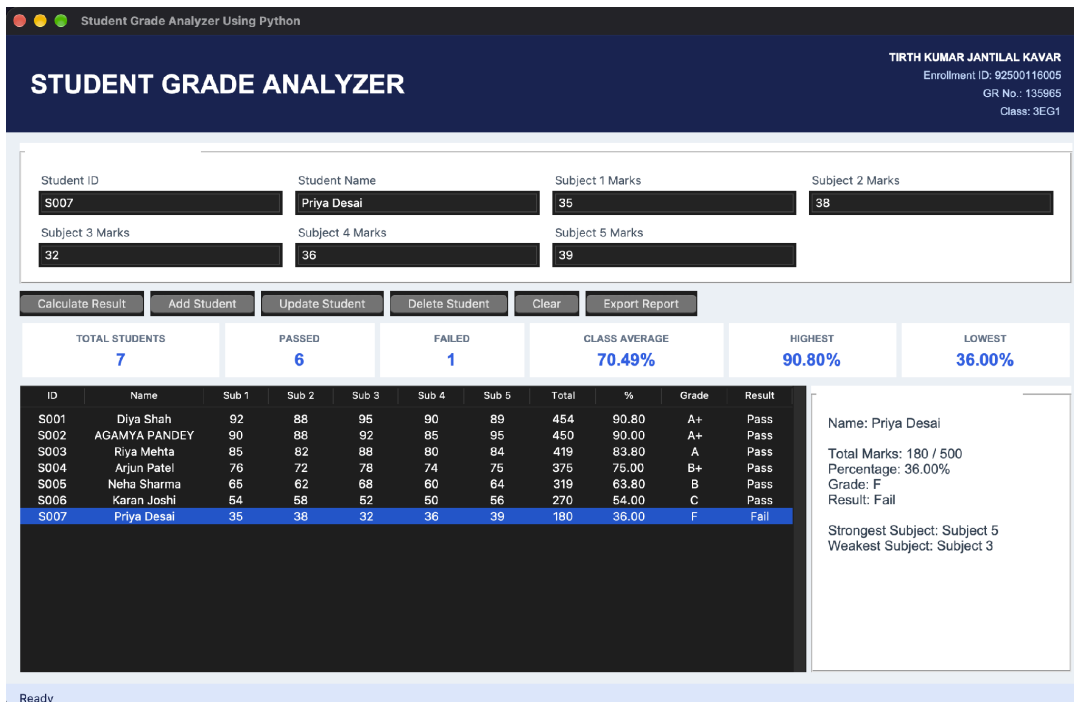


Figure 3.4 Class statistics dashboard and individual subject report

GUI Output Screenshots

INVENTORY TRACKER

TIRTH KUMAR JANTIL KAVAR
Enrollment ID: B2500116005
GR No.: 130985
Class: 3EG1

Item Details

Item ID:
Unit Price:

Item Name:
Supplier:

Category:
Reorder Level:

Quantity:

Add Item Update Item Delete Item Clear Export CSV Search: Search Item Show All

TOTAL ITEMS: **15**
TOTAL QUANTITY: **172**
LOW STOCK: **5**
OUT OF STOCK: **3**
INVENTORY VALUE: **₹306,765.00**

Item ID	Item Name	Category	Quantity	Unit Price	Supplier	Reorder Level	Stock Value	Status
I001	Wireless Mouse	Electronics	25	650.00	Tech Supplies	10	16250.00	IN STOCK
I002	Printer Paper	Stationery	5	280.00	Office Mart	8	1400.00	LOW STOCK
I003	USB Cable	Electronics	0	150.00	Tech Supplies	5	0.00	OUT OF STOCK
I004	Mechanical Keyboard	Electronics	18	2450.00	Digital World	6	44100.00	IN STOCK
I005	A4 Notebook	Stationery	7	95.00	Office Mart	10	665.00	LOW STOCK
I006	Office Chair	Furniture	12	4850.00	Comfort Furnishings	4	58200.00	IN STOCK
I007	Whiteboard Marker	Stationery	0	45.00	Campus Store	15	0.00	OUT OF STOCK
I008	LED Monitor	Electronics	9	12750.00	Digital World	3	114750.00	IN STOCK
I009	Desk Lamp	Furniture	4	1250.00	Comfort Furnishings	5	5000.00	LOW STOCK
I010	Network Router	Networking	14	3200.00	Net Solutions	5	44800.00	IN STOCK
I011	Ethernet Cable	Networking	30	275.00	Net Solutions	12	8250.00	IN STOCK
I012	Laser Toner	Printing	2	3800.00	Print House	4	7600.00	LOW STOCK
I013	Document Scanner	Electronics	0	18500.00	Digital World	2	0.00	OUT OF STOCK

Ready

Figure 4.1 Inventory management interface and item details form

INVENTORY TRACKER

TIRTH KUMAR JANTIL KAVAR
Enrollment ID: B2500116005
GR No.: 130985
Class: 3EG1

Item Details

Item ID:
Unit Price:

Item Name:
Supplier:

Category:
Reorder Level:

Quantity:

Add Item Update Item Delete Item Clear Export CSV Search: Search Item Show All

TOTAL ITEMS: **16**
TOTAL QUANTITY: **212**
LOW STOCK: **5**
OUT OF STOCK: **3**
INVENTORY VALUE: **₹308,165.00**

Item ID	Item Name	Category	Quantity	Unit Price	Supplier	Reorder Level	Stock Value	Status
I004	Mechanical Keyboard	Electronics	18	2450.00	Digital World	6	44100.00	IN STOCK
I005	A4 Notebook	Stationery	7	95.00	Office Mart	10	665.00	LOW STOCK
I006	Office Chair	Furniture	12	4850.00	Comfort Furnishings	4	58200.00	IN STOCK
I007	Whiteboard Marker	Stationery	0	45.00	Campus Store	15	0.00	OUT OF STOCK
I008	LED Monitor	Electronics	9	12750.00	Digital World	3	114750.00	IN STOCK
I009	Desk Lamp	Furniture	4	1250.00	Comfort Furnishings	5	5000.00	LOW STOCK
I010	Network Router	Networking	14	3200.00	Net Solutions	5	44800.00	IN STOCK
I011	Ethernet Cable	Networking	30	275.00	Net Solutions	12	8250.00	IN STOCK
I012	Laser Toner	Printing	2	3800.00	Print House	4	7600.00	LOW STOCK
I013	Document Scanner	Electronics	0	18500.00	Digital World	2	0.00	OUT OF STOCK
I014	File Folder	Stationery	40	35.00	Campus Store	15	1400.00	IN STOCK
I015	Extension Board	Electrical	6	725.00	Power Point	8	4350.00	LOW STOCK
I016	File Folder	Stationery	40	35.00	Campus Store	15	1400.00	IN STOCK

Item added

Figure 4.2 Stock item entry, category selection, and reorder levels

INVENTORY TRACKER

TIRTH KUMAR JANTILAL KAVAR

Enrollment ID: 92550116006

OR No.: 135985

Class: 9EG1

Item Details

Item ID

Item Name

Category

Quantity

Unit Price

Supplier

Reorder Level

Add Item

Update Item

Delete Item

Clear

Export CSV

Search: 1001

Search Item

Show All

TOTAL ITEMS

TOTAL QUANTITY

LOW STOCK

OUT OF STOCK

INVENTORY VALUE

16

275

4

3

₹314,150.00

Item ID	Item Name	Category	Quantity	Unit Price	Supplier	Reorder Level	Stock Value	Status
1001	Wireless Mouse	Electronics	25	650.00	Tech Supplies	10	16250.00	IN STOCK

Showing 1 matching item(s)

Figure 4.3 Color-coded inventory table showing stock status

TOTAL ITEMS	TOTAL QUANTITY	LOW STOCK	OUT OF STOCK	INVENTORY VALUE
16	275	4	3	₹314,150.00

Figure 4.4 Inventory metrics and stock valuation summary bar

GUI Output Screenshots

LOG FILE ANALYZER

TIRTH KUMAR JANTIL KAVAR

Enrollment ID: 92500116005

GR No.: 135985

Class: 3EG1

Log File

Browse File

Analyze Logs

Export CSV

TOTAL LOGS

33

DEBUG

5

INFO

14

WARNING

6

ERROR

6

CRITICAL

2

Level: All

Search message:

Log Entries

Date	Time	Level	Message
2026-09-25	08:00:01	INFO	Application service started successfully
2026-09-25	08:00:02	DEBUG	Configuration file loaded from production environment
2026-09-25	08:00:03	DEBUG	Database connection pool initialized with 20 connections
2026-09-25	08:01:15	INFO	User authentication service is ready
2026-09-25	08:05:42	INFO	User 1045 logged in successfully
2026-09-25	08:08:09	WARNING	Memory usage reached 78 percent
2026-09-25	08:10:33	DEBUG	Scheduled cache cleanup started

Errors & Critical Events

Date	Time	Level	Message
2026-09-25	08:15:17	ERROR	Database connection timed out
2026-09-25	08:25:31	ERROR	Payment gateway returned status 503
2026-09-25	08:25:34	ERROR	Payment gateway returned status 503
2026-09-25	08:40:11	ERROR	Unable to write temporary export file
2026-09-25	08:45:36	CRITICAL	Primary server became unavailable
2026-09-25	08:50:14	ERROR	Database connection timed out
2026-09-25	09:02:16	CRITICAL	Backup storage is not responding

Total: 33 Errors: 6 Warnings: 6 Critical: 2 Error percentage: 24.24%

Most common error message: Database connection timed out

Figure 5.1 Log analyzer interface and file selection dialog

LOG FILE ANALYZER

TIRTH KUMAR JANTIL KAVAR

Enrollment ID: 92500116005

GR No.: 135985

Class: 3EG1

Log File

Browse File

Analyze Logs

Export CSV

TOTAL LOGS

33

DEBUG

5

INFO

14

WARNING

6

ERROR

6

CRITICAL

2

Level: ERROR

Search message:

Log Entries

Date	Time	Level	Message
2026-09-25	08:15:17	ERROR	Database connection timed out
2026-09-25	08:25:31	ERROR	Payment gateway returned status 503
2026-09-25	08:25:34	ERROR	Payment gateway returned status 503
2026-09-25	08:40:11	ERROR	Unable to write temporary export file
2026-09-25	08:50:14	ERROR	Database connection timed out
2026-09-25	09:02:20	ERROR	Scheduled backup failed

Errors & Critical Events

Date	Time	Level	Message
2026-09-25	08:15:17	ERROR	Database connection timed out
2026-09-25	08:25:31	ERROR	Payment gateway returned status 503
2026-09-25	08:25:34	ERROR	Payment gateway returned status 503
2026-09-25	08:40:11	ERROR	Unable to write temporary export file
2026-09-25	08:45:36	CRITICAL	Primary server became unavailable
2026-09-25	08:50:14	ERROR	Database connection timed out
2026-09-25	09:02:16	CRITICAL	Backup storage is not responding

Total: 33 Errors: 6 Warnings: 6 Critical: 2 Error percentage: 24.24%

Most common error message: Database connection timed out

Figure 5.2 Parsed server logs categorized by severity level

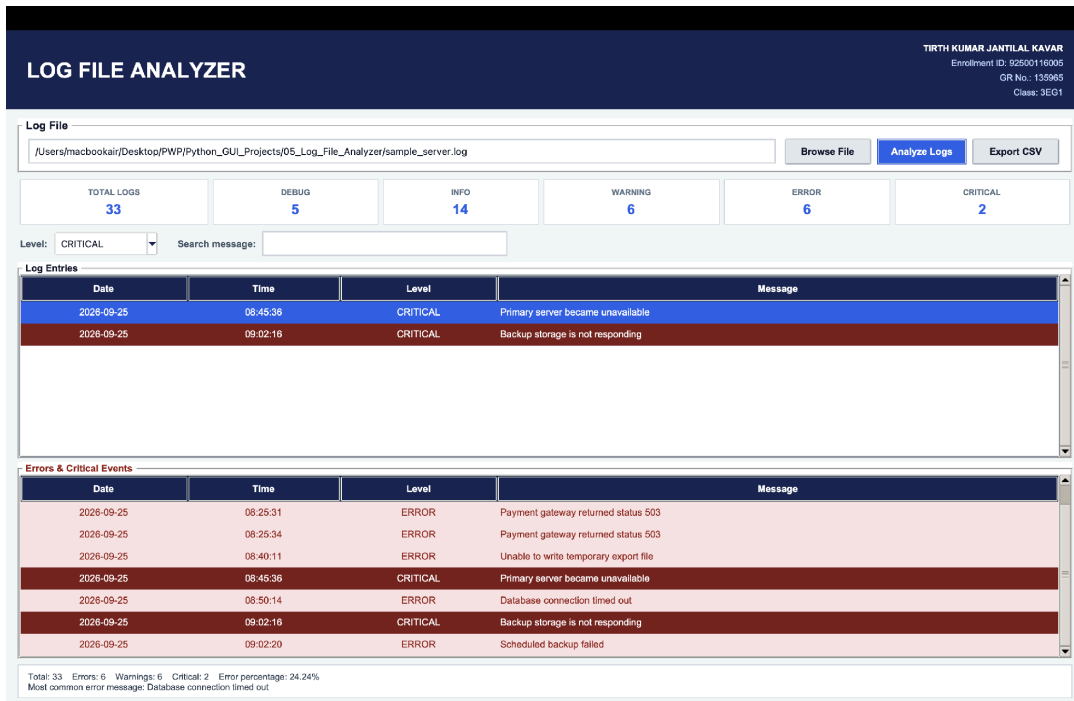


Figure 5.3 Filtered error and critical events table

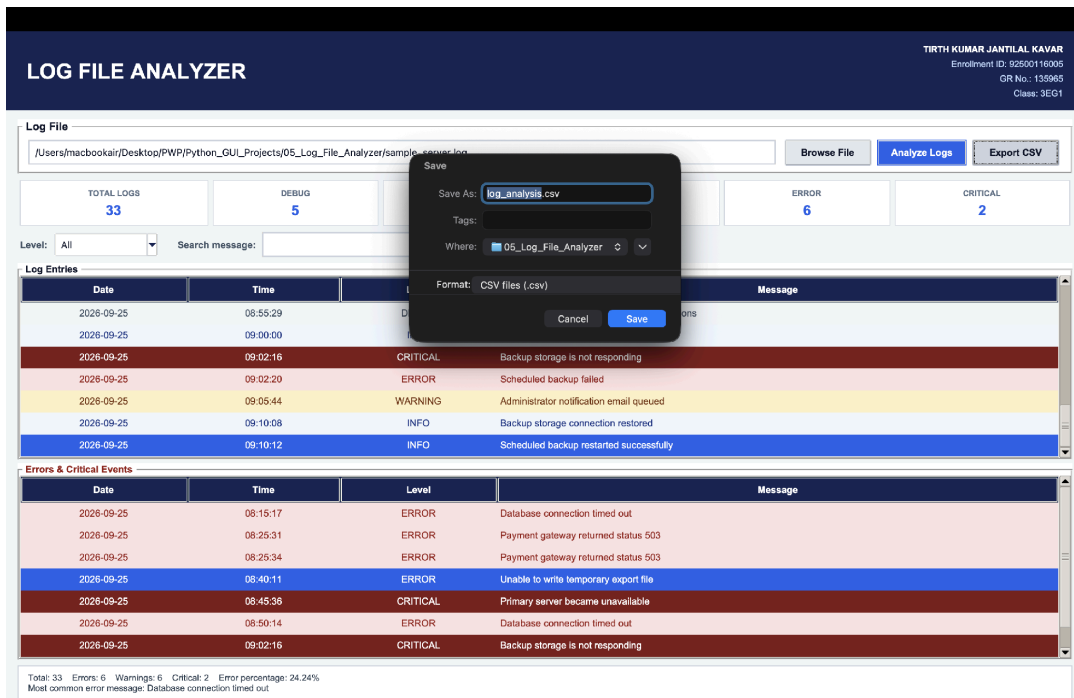


Figure 5.4 Summary metrics, error breakdown, and CSV export